# Computational Superconductivity: A Study Plan

*Calibrated for a BS in physics, ten years cold.*

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## What this assumes about you

You once knew quantum mechanics, statistical mechanics, and electromagnetism well enough to pass exams on them. You may or may not have taken a solid-state course. You almost certainly never ran a density functional theory calculation, because that's usually a graduate skill.

The good news: most of the vocabulary that makes computational superconductivity papers look impenetrable is built on machinery you already learned, wearing notation you haven't seen. Reciprocal space, perturbation theory, normal modes, Fermi statistics — it's all in there. The rust is real, but it's surface rust.

**Budget:** about 12 weeks at 5–8 hours per week. Phases 2 through 4 are the actual content; everything before is scaffolding and everything after is optional depth.

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## Week 0 — Diagnostic

Answer these honestly before deciding how much of Phase 1 you need.

1. What does a wavevector **k** actually label, and why is it confined to the first Brillouin zone?
2. Why does a periodic potential produce energy bands? (If "Bloch's theorem" surfaces, you're fine.)
3. What is a Fermi surface, and why do only states near it matter for most low-energy physics?
4. What is a phonon? What distinguishes an acoustic branch from an optical one?
5. Why does the density of states at the Fermi level, N(E_F), show up in so many expressions?
6. What do $a^\dagger$ and $a$ do in second-quantized notation?

**Four or more confident answers:** skim Phase 1, spend a week, move on. **Fewer than four:** give Phase 1 its full three weeks. This is not a setback — it's the cheapest possible time to fix it.

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## Phase 1 — Solid-state refresher (1–3 weeks)

**Why it matters:** Every quantity downstream is an integral over the Brillouin zone or a sum over bands. If reciprocal space feels shaky, nothing later will land.

**Terms:** Bloch's theorem, reciprocal lattice, Brillouin zone, band structure, band gap, density of states, Fermi energy, Fermi surface, effective mass, phonon, acoustic and optical branches, dispersion relation, Debye model, Debye frequency.

**Do this:** Read Steven Simon, *The Oxford Solid State Basics*, chapters on periodic solids, band structure, and phonons. It's the single best rust-remover in the field — written for undergraduates, brisk, and honest about what it's skipping.

**Self-test:** Sketch the free-electron band structure of a simple cubic metal and mark where a gap would open. Explain out loud why a half-filled band gives a metal.

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## Phase 2 — Superconductivity itself (2 weeks)

**Why it matters:** You need to know what you're trying to predict before you can follow how it's predicted.

**Terms:** Cooper pair, BCS theory, energy gap Δ, coherence length ξ, London penetration depth, Meissner effect, type I vs type II, critical temperature/field/current, isotope effect, weak vs strong coupling, conventional vs unconventional pairing.

**Do this:** Tinkham, *Introduction to Superconductivity*, chapters 1–3. Read for the physical picture, not the derivations. The one derivation worth following carefully is the BCS gap equation, because its structure — a self-consistent integral over states near E_F, weighted by an interaction — is the template for everything in Phase 5.

**The key insight to carry forward:** BCS says an arbitrarily weak attraction destabilizes the normal Fermi sea. The attraction comes from lattice distortion — one electron polarizes the ions, the second electron feels that polarization. Everything computational downstream is an attempt to calculate the strength of exactly that effect from first principles.

**Self-test:** Explain the isotope effect and why it was historical evidence that phonons mediate pairing.

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## Phase 3 — Electronic structure and DFT (3 weeks)

**Why it matters:** This is the genuinely new material. It's also where most of the acronyms live.

**Terms:** density functional theory, Hohenberg–Kohn theorems, Kohn–Sham equations, exchange–correlation functional, LDA, GGA/PBE, pseudopotential, PAW, plane-wave cutoff, k-point mesh, Monkhorst–Pack grid, smearing (Methfessel–Paxton, Marzari–Vanderbilt), self-consistent field convergence, N(E_F).

**Do this:** Two tracks in parallel.

*Conceptual:* Richard Martin, *Electronic Structure*, or David Sholl and Janice Steckel, *Density Functional Theory: A Practical Introduction* — the latter is friendlier and faster if you want to be running calculations sooner.

*Practical:* Install Quantum ESPRESSO and work the introductory tutorials. Compute the band structure and density of states of aluminum or $MgB_2$. This is not optional busywork — the parameters you'll be reading about in papers (cutoffs, k-meshes, smearing widths) only become real once you've watched a calculation fail to converge because you set one badly.

**Self-test:** Explain what a pseudopotential replaces and why that replacement is safe for valence physics.

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## Phase 4 — Lattice dynamics (2 weeks)

**Why it matters:** Phonons are half of the electron–phonon coupling, and phonon calculations are where predicted materials most often fall apart.

**Terms:** dynamical matrix, density functional perturbation theory (DFPT), finite-displacement/supercell method, q-point mesh, phonon dispersion, soft modes, imaginary frequencies, dynamical stability, anharmonicity, SSCHA, zero-point energy.

**Do this:** Read the DFPT sections of the Quantum ESPRESSO documentation and run a phonon calculation on the structure you converged in Phase 3.

**The key idea:** Imaginary phonon frequencies mean the structure is sitting at a saddle point, not a minimum — it would spontaneously distort. A large fraction of the hydride literature consists of arguments about whether a computationally predicted structure is *actually* dynamically stable once anharmonicity and zero-point motion are handled properly. Hydrogen is light, so those corrections are enormous.

**Self-test:** Explain why a soft mode near a particular q-vector can signal both a structural instability and strong electron–phonon coupling.

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## Phase 5 — The coupling (3 weeks — the core)

**Why it matters:** This is what the entire field is about. Everything before was preparation.

**Terms:** electron–phonon matrix element g, Eliashberg spectral function $\alpha^2F(\omega)$, coupling constant $\lambda$, logarithmic average frequency $\omega$_log, second moment $\omega_2$, phonon linewidth $\gamma$, Fermi surface nesting, Wannier functions, Wannier interpolation, EPW.

**Do this:** Feliciano Giustino, "Electron-phonon interactions from first principles," *Reviews of Modern Physics* 89, 015003 (2017). Dense, but it is the definitive modern treatment and it is written to be read.

**The insight that unlocks the notation:** $\alpha^2F(\omega)$ is a phonon density of states, weighted by how strongly each phonon couples to electrons at the Fermi surface. And $\lambda$ is simply its first inverse moment — twice the integral of $\alpha^2F(\omega)/\omega$. Once you see that $\lambda$ is *just a number* *extracted from a curve*, the symbol stops being intimidating.

Rules of thumb worth memorizing: λ below roughly 0.5 is weak coupling, 0.5–1.0 is intermediate, above 1.5 is strong coupling where simple formulas start to fail.

**Self-test:** Explain why low-frequency phonons contribute disproportionately to λ, and why that creates a tension with achieving high Tc.

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## Phase 6 — Getting to Tc (2 weeks)

**Why it matters:** This is the payoff, and also where the honest caveats live.

**Terms:** McMillan formula, Allen–Dynes modification, strong-coupling correction factors $f_1$ and $f_2$, Coulomb pseudopotential μ*, Migdal–Eliashberg theory, isotropic vs anisotropic Eliashberg equations, Migdal's theorem, SCDFT.

**Do this:** Allen and Mitrović, "Theory of Superconducting Tc," *Solid State Physics* vol. 37 (1982). It's old, and nearly everything downstream cites it.

**The honest caveat to internalize:** μ*, the parameter encoding Coulomb repulsion between the pairing electrons, is usually not calculated. It's set to 0.1 or 0.13 because those values reproduce known materials. Tc depends on it sensitively. When a paper reports "predicted Tc = 200 K," part of that number is a fitted constant. Superconducting DFT exists partly to remove this fudge, which is why it keeps appearing in the literature despite being harder to run.

**Self-test:** Explain what Migdal's theorem assumes and why it becomes questionable at very strong coupling.

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## Phase 7 — Current frontier (ongoing)

**Terms:** high-pressure hydrides ($H_3S$, $LaH_{10}$, $YH_9$), crystal structure prediction (USPEX, CALYPSO, AIRSS), convex hull, formation enthalpy, machine-learning screening, and on the unconventional side: spin fluctuations, RPA, FLEX, DMFT, d-wave pairing, cuprates, nickelates.

This area moves quickly and my knowledge has a cutoff, so use recent review articles rather than anything secondhand. Be aware that several high-profile experimental claims of near-ambient superconductivity have been contested or retracted; check the current status of any specific claim before building on it.

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## Notation traps

These cause more confusion than any of the physics.

| Symbol | Trap |
|---|---|
| $\lambda$ | Penetration depth in classic superconductivity texts; electron–phonon coupling constant in computational papers. Completely unrelated quantities. |
| $\mu$ vs $\mu^*$ | Chemical potential vs Coulomb pseudopotential. The asterisk is doing enormous work. |
| $\alpha^2F(\omega)$ | Not "alpha squared times F." A single named object, the Eliashberg spectral function. Read it as one word. |
| $N(E_F)$ | Sometimes per spin, sometimes both spins. The resulting factor of 2 propagates into $\lambda$. Check conventions. |
| k vs q | Electron wavevector vs phonon wavevector. Matrix elements couple both, and sloppy writing conflates them. |
| $\Delta$ | Superconducting gap, but also used for finite differences and Laplacians in the same papers. |
| $\omega_{log}$, $\omega_2$ | Different weighted averages of the same phonon spectrum, appearing in different Tc formulas. |

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## Practical advice

**Keep a pipeline glossary, not an alphabetical one.** For each term, record three things: what it is, what it's computed *from*, and what it feeds *into*. The structure of the pipeline is what makes the vocabulary stick — isolated definitions evaporate.

**Run calculations earlier than feels comfortable.** Reading about k-point convergence teaches you less in three hours than one afternoon of watching a calculation give a wrong answer because your mesh was too coarse.

**Don't chase the frontier first.** The hydride papers are the exciting ones, and they're the least readable without Phases 3 through 6. They will still be there in ten weeks.

**Expect Phase 3 to be the hardest.** Not because DFT is conceptually difficult, but because it's genuinely new material and the software has a learning curve. Budget the frustration.

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## Core reading list

| Phase | Source |
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| 1 | Simon, *The Oxford Solid State Basics* |
| 2 | Tinkham, *Introduction to Superconductivity*, ch. 1–3 |
| 3 | Sholl & Steckel, *DFT: A Practical Introduction*; Quantum ESPRESSO tutorials |
| 4 | Quantum ESPRESSO DFPT documentation |
| 5 | Giustino, *Rev. Mod. Phys.* 89, 015003 (2017) |
| 6 | Allen & Mitrović, *Solid State Physics* 37 (1982) |
| 7 | Recent review articles — check publication dates |